STAT201 · STATISTICS · LECTURE 7

Sampling Distributions and the Central Limit Theorem

Why the sample mean fluctuates, how to measure this fluctuation, and why it is almost always normal

Syllabus topic 7 · week 7 · 4 academic hours

LECTURE GOALS

- distinguish a population from a sample, a parameter from a statistic;
- explain that the sample mean is a random variable and describe its distribution (expectation and standard error);
- state the central limit theorem (CLT) and the conditions for its use;
- find probabilities for $\bar{X}$ and $\hat{p}$ using the normal distribution;
- determine the sample size from a required precision and understand why $n-1$ is used in the sample variance.

1 Population, sample, statistic

The *population* is all the objects we want to learn about (all bank customers, all parts of a batch). The *sample* is its part that we actually observe. A *parameter* is a number characterizing the population (μ, σ, p); it is fixed and usually unknown. A *statistic* is a number computed from a sample ($\bar{x}$, s, $\hat{p}$); it is known but *changes from sample to sample*.

For conclusions to be reliable the sample must be random (each object has a known nonzero chance of entering it). A simple random sample of n independent observations is denoted $X_1, \dots, X_n$. A non-representative sample (for example, a phone survey during working hours only) gives biased estimates that cannot be fixed by increasing n.

SAMPLING DISTRIBUTION

The sampling distribution of a statistic is the distribution of the values it takes over all possible samples of size n.

2 The sample mean

Let $X_1, \dots, X_n$ be independent, $\mathrm{E}[X_i] = \mu$, $\mathrm{Var}(X_i) = \sigma^2$. The mean is $\bar{X} = \frac{1}{n}\sum X_i$. From the properties of Lecture 4:

$$\mathrm{E}\big[\bar{X}\big] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\bar{X}) = \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}. \tag{2.1}$$

The first equality means that $\bar{X}$ is an *unbiased* estimator of μ: averaged over all samples it hits the target. The second: the spread decreases as n grows, but slowly — to halve the error one needs a sample four times larger. The standard deviation of a statistic is called the *standard error* (SE).

COMMON MISTAKE

σ is the spread of individual observations, $\text{SE} = \sigma/\sqrt{n}$ is the spread of the mean. These are different numbers: in problems about "the sample mean" divide by $\sqrt{n}$, in problems about "a single observation" do not.

EXAMPLE 1. A SAMPLING DISTRIBUTION IN A SMALL EXAMPLE

The population is $\{1, 2, 3\}$, $\mu = 2$, $\sigma^2 = 2/3$. Take samples of 2 elements with replacement (9 equally likely pairs). The mean $\bar{x}$ takes the values $1; 1.5; 2; 2.5; 3$ with probabilities $\frac{1}{9}, \frac{2}{9}, \frac{3}{9}, \frac{2}{9}, \frac{1}{9}$. The expectation is $\mathrm{E}\big[\bar{X}\big] = 2 = \mu$, the variance $\mathrm{Var}(\bar{X}) = \frac{1}{3} = \frac{\sigma^2}{n} = \frac{2/3}{2}$ — formula (2.1) is confirmed by direct enumeration.

If the sample is drawn without replacement from a finite population of size N and makes up a noticeable share ($n/N > 5\%$), the standard error is multiplied by the correction $\sqrt{\frac{N-n}{N-1}}$. For example, $N = 500$, $n = 100$, $\sigma = 4$: $\text{SE} = 0.4 \cdot 0.895 = 0.358$ instead of 0.4.

3 The central limit theorem

CLT

Let $X_1, \dots, X_n$ be independent identically distributed variables with finite μ and σ^2. Then for large n the distribution of $\bar{X}$ is approximately normal:

$$\bar{X} \;\approx\; N\!\left(\mu,\, \frac{\sigma^2}{n}\right), \qquad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0,1), \tag{3.1}$$

whatever the original distribution. Similarly for the sum: $\sum X_i \approx N(n\mu,\, n\sigma^2)$.

Rule of thumb: $n \geq 30$ is enough for moderately skewed distributions; for strongly skewed ones or ones with outliers more is needed; if the original distribution is normal, $\bar{X}$ is normal for any n.

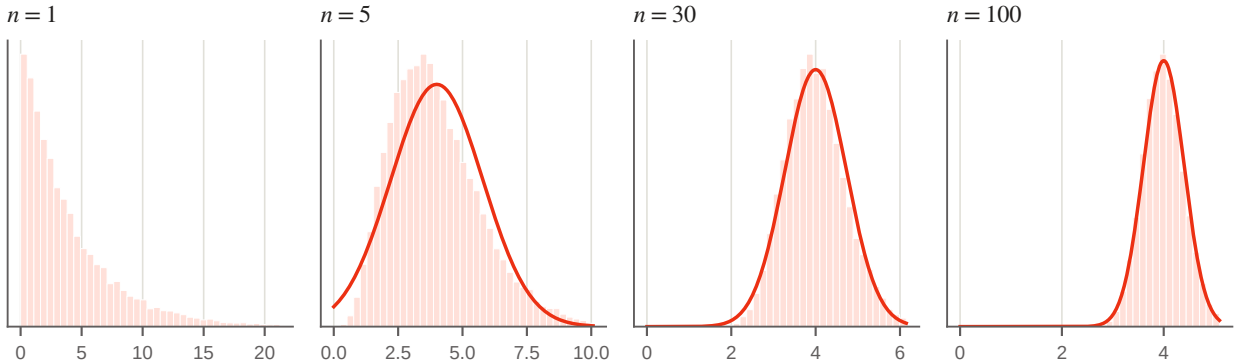


Fig. 1. Simulation: the distribution of the sample mean from an exponential population ($\mu = \sigma = 4$) over 20 000 samples. For $n = 1$ strongly skewed; for $n = 30$ almost bell-shaped; the red line is the normal approximation $N(4, (4/\sqrt{n})^2)$.

EXAMPLE 2. PROCESSING TIME OF AN APPLICATION

The processing time of a loan application: $\mu = 12$ days, $\sigma = 4$. A sample of $n = 36$ applications. SE $= 4/6 = 0.667$. $P(\bar{X} > 13) = P(Z > 1.5) = 0.0668$. $P(11 < \bar{X} < 13) = P(-1.5 < Z < 1.5) = 0.8664$. The mean of 36 applications lies in 95% of cases within $12 \pm 1.96 \cdot 0.667$, i.e. from 10.69 to 13.31 days. Note: for a single application $P(X > 13)$ is substantially larger (about 40% for a normal X).

EXAMPLE 3. A SKEWED POPULATION

Waiting time is exponential with a mean of 4 minutes. $n = 40$, SE $= 4/\sqrt{40} = 0.632$. $P(\bar{X} > 5) \approx P(Z > 1.58) = 0.0569$. The exact value (a sum of exponential variables has a gamma distribution) is 0.0646; the discrepancy remains because of the noticeable skewness, but the order of magnitude is right.

4 The sample proportion

For a proportion $\hat{p} = X/n$, where $X \sim \text{Bin}(n, p)$ is the number of "successes" in the sample, by the CLT (normal approximation to the binomial, Lecture 5):

$$\text{E}[\hat{p}] = p, \qquad \text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}, \qquad \hat{p} \approx N\left(p,\ \frac{p(1-p)}{n}\right) \text{ when } np \geq 10,\ n(1-p) \geq 10. \tag{4.1}$$

EXAMPLE 4. CUSTOMER SATISFACTION

Among bank customers 40% are satisfied with the service. $n = 100$ were surveyed. SE $= \sqrt{0.4 \cdot 0.6/100} = 0.049$. $P(\hat{p} > 0.50) = P(Z > 2.04) = 0.0206$. $P(0.35 < \hat{p} < 0.45) = P(|Z| < 1.02) = 0.6926$: two thirds of samples give a proportion within ± 5 percentage points of the true one.

5 Sample size and precision

Suppose we need the sample mean to differ from μ by no more than E with probability 95%. The condition $1.96\,\sigma/\sqrt{n} \leq E$ gives

$$n \geq \left(\frac{1.96\,\sigma}{E}\right)^2. \tag{5.1}$$

For $\sigma = 4$ days and $E = 1$ day: $n \geq (7.84)^2 = 61.5$, i.e. $n = 62$. For proportions one uses $n \geq \dfrac{1.96^2\,p(1-p)}{E^2}$; when p is unknown the worst case $p = 0.5$ is taken: for $E = 0.03$ (3 percentage points) we get $n \geq 1067.1$, i.e. 1068 respondents. This number explains why opinion polls usually contain about a thousand people.

6 The sample variance and division by $n-1$

The sample variance $s^2 = \dfrac{\sum(X_i - \bar{X})^2}{n-1}$ is an unbiased estimator of σ^2: $\mathrm{E}[s^2] = \sigma^2$. If we divide by n, the estimate is systematically too low, because deviations are computed from $\bar{X}$ rather than from the unknown μ: the sample is "closer to its own mean" than to μ.

EXAMPLE 5. A CHECK ON A SMALL POPULATION

In the example above ($\{1, 2, 3\}$, samples of 2 with replacement) $\sigma^2 = 0.667$. The average value of s^2 with the divisor $n-1=1$ is $0.667 = \sigma^2$, and with the divisor $n=2$ it is 0.333, half as much.

REMEMBER

The sampling distribution of s^2 (more precisely, of $(n-1)s^2/\sigma^2$) for a normal population is chi-squared with $n-1$ degrees of freedom. It will be needed in Lecture 11.

7 Key terms

Term	*In Russian*	Meaning
Population	*Генеральная совокупность*	all objects of the study
Sample	*Выборка*	the observed part of the population
Parameter / statistic	*Параметр / статистика*	μ, σ, p / $\bar{x}, s, \hat{p}$
Sampling distribution	*Выборочное распределение*	the distribution of a statistic over all samples
Standard error	*Стандартная ошибка*	$\sigma/\sqrt{n}$; $\sqrt{p(1-p)/n}$
Unbiasedness	*Несмещённость*	$\mathrm{E}[\text{estimator}] = \text{parameter}$
Central limit theorem (CLT)	*ЦПТ*	$\bar{X} \approx N(\mu, \sigma^2/n)$

Formulas of Lecture 7

$$\mathrm{E}[\bar{X}] = \mu, \quad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}, \quad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0,1);$$

$$\mathrm{E}[\hat{p}] = p, \quad \mathrm{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}};$$

$$n \geq \left(\frac{1.96\sigma}{E}\right)^2, \qquad n \geq \frac{1.96^2 p(1-p)}{E^2}; \qquad s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}.$$

8 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] The average bill in a chain of cafés is $\mu = 50$ thousand UZS, $\sigma = 12$. A sample of $n = 36$ bills is taken.

(a) Find the expectation and standard error of $\bar{X}$.
(b) Find $P(\bar{X} > 53)$ and $P(48 < \bar{X} < 52)$.

Problem 2 [A] $\sigma = 20$. Find the standard error of the mean for $n = 25$ and $n = 100$. What sample size is needed for SE $= 1$?

Problem 3 [A] The share of customers with the mobile app is $p = 0.30$. $n = 200$ were surveyed. Find SE$(\hat{p})$, $P(\hat{p} > 0.35)$ and $P(0.28 < \hat{p} < 0.32)$.

Problem 4 [B] The size of a deposit is strongly right-skewed: $\mu = 8$ million, $\sigma = 5$ million. A sample of $n = 50$ deposits.

(a) May the normal distribution be used for $\bar{X}$? Justify.
(b) Find $P(\bar{X} < 7)$ and $P(\bar{X} > 9.5)$.
(c) Can $P(X < 7)$ be found in the same way for a single deposit?

Problem 5 [B] Each of 100 transactions has $\mu = 2$ million UZS and $\sigma = 1.5$ million, the transactions are independent. Find the approximate probability that the total volume exceeds 230 million.

Problem 6 [B] We need to estimate the average time to resolve a request ($\sigma \approx 15$ minutes) to within $E = 2$ minutes at 95% confidence; then to estimate the share of satisfied customers to within ± 0.04.

(a) Find the required n for the mean.
(b) How does n change if the required precision is $E = 1$ minute?
(c) Find n for the proportion with unknown p and with $p \approx 0.2$.

Problem 7 [B] The defect rate on a line is $p = 0.05$. A batch of $n = 400$ parts is inspected.

(a) Check the conditions of the normal approximation and find SE$(\hat{p})$.
(b) Find the probability that the sample defect rate exceeds 7%.
(c) Find the probability that the sample defect rate turns out below 3%.

Problem 8 [B] The population is $\{2, 4, 6\}$; samples of 2 with replacement are taken (9 equally likely).

(a) Find μ and σ^2.
(b) Write the distribution of $\bar{X}$ and check $\mathrm{E}[\bar{X}] = \mu$, $\mathrm{Var}(\bar{X}) = \sigma^2/2$.
(c) Find the mean value of s^2 (with divisor $n - 1$).

Problem 9 [B] From a population of $N = 400$ customers with $\sigma = 20$ a sample of $n = 100$ is drawn without replacement. Find the standard error of the mean with and without the finite population correction.

Problem 10 [C] Derive $\mathrm{Var}(\bar{X}) = \sigma^2/n$ and show that $\mathrm{E}[s^2] = \sigma^2$ if $s^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2$.

Problem 11 [C] The law of large numbers. For $\sigma = 4$ and $\varepsilon = 1$ Chebyshev's inequality gives $P(|\bar{X} - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$. Estimate the probability for $n = 100$ and compare with the normal approximation. What does the probability tend to as $n \to \infty$?

Problem 12 [C] An online poll in the bank's Telegram channel showed 80% satisfied customers with $n = 5000$. Suppose $\mathrm{SE} \approx 0.006$. Does this mean that the share of satisfied customers among all customers lies within $0.80 \pm 2 \cdot 0.006$? Explain.